LESSON 4 • TERM 1 • LEVEL 1

# 100 Square Explorer

Whole numbers: patterns in rows and columns

**TODAY WE WILL**

Explore the 100 square and find the patterns hiding in its rows and columns.



# Get ready

0–4 MIN

- Your 100 square (the kit page)
- About 12 counters, all one kind
- Two or three colours
- Workbook, pencil and paper

**Counters stay on the table, never near the mouth or ears.**

4–10 MIN

# Warm-up: count in tens

Show both hands for every ten

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 |

Now count back from 100. Whisper, and shout on 50!




Count along with me!

4–10 MIN

# Fives and twos

One hand is five fingers. Count in fives:

5 10 15 20 25 30

Socks come in pairs. A pair is two. Count in twos:

2 4 6 8 10 12

4–10 MIN

# What comes next?

10 20 30 ___

5 10 15 ___

2 4 6 ___

28 29 ___

38 39 ___

Say the next number. Show it on your fingers if you like.

# Sneak-peek puzzles

10–14 MIN

**Ideas we have not learnt yet. A guess is a great start. Nobody is marked.**

## PUZZLE 1: anyone can try

Count in 5s. What are the next two numbers?

| 5 | 10 | 15 | 20 | ___ | ___ |
|---|---|---|---|---|---|

## PUZZLE 2: think

This is part of the 100 square. A counter is on 36.

Ravi moves it down one square.

Which number is it on now? Write how you know.

| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
|---|---|---|---|---|---|---|---|---|---|
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| | | | | | | | | | |

# Explore: read the square

14–28 MIN

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

1. Put a finger on 1. Slide it along to 10.
2. Where does 11 sit? Slide your finger down.
3. Find 100. Which corner is it in?

14–28 MIN

# Explore: cover the tens

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

1. Count in tens: 10, 20, 30, ...
2. Put a counter on each number you say.
3. What do you see?

SAY IT

The counters make a line that goes ___ (across or down).

# Explore: fives and twos

14–28 MIN

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |

1. Clear your counters. Count in fives to 50.
2. What do you see?
3. Clear again. Count in twos to 20.
4. Which numbers have no counter?

**Counting in 2s: will 37 get a counter? Will 60?**

14–28 MIN

# Explore: move the counter

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |

1. Counter on 34. Move it right 1.
2. Now down 1. Down 1 again.
3. Left 1. Then up 1.

WHAT DO YOU NOTICE?

Which way adds 1?

Which way adds 10?

28–38 MIN

# Across: 1 more, 1 less

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |







**Right: 1 more. Left: 1 less.**

At the end of a row, the next number starts the next row: 39, 40, 41.

28–38 MIN

# Down: 10 more, up: 10 less

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |




**Down: 10 more. Up: 10 less.**

A row has ten squares. So one step down is ten more.

28–38 MIN

# Your turn: write the number

Write on your paper. Timer: 1 minute

1

10 more than 36

2

10 less than 58

3

1 more than 29

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

28–38 MIN

# Check: write the number

1

10 more than 36

**46**

2

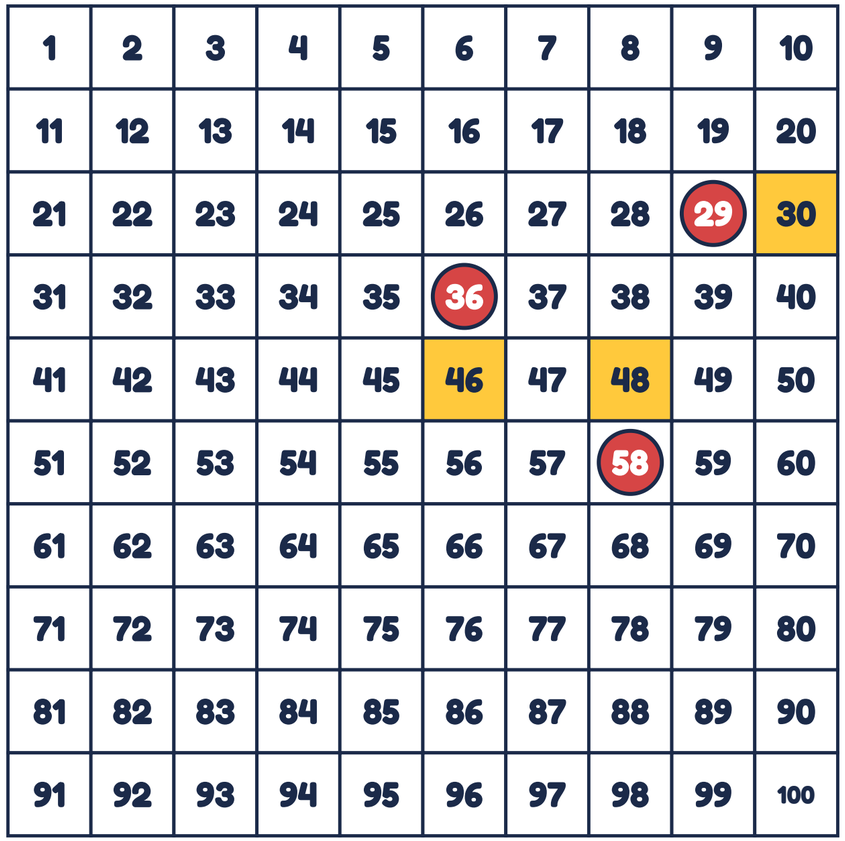
10 less than 58

**48**

3

1 more than 29

**30**



| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

# Stripes and last digits

28–38 MIN

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

10s: one column

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

5s: two columns

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

2s: stripes

THE LAST DIGIT

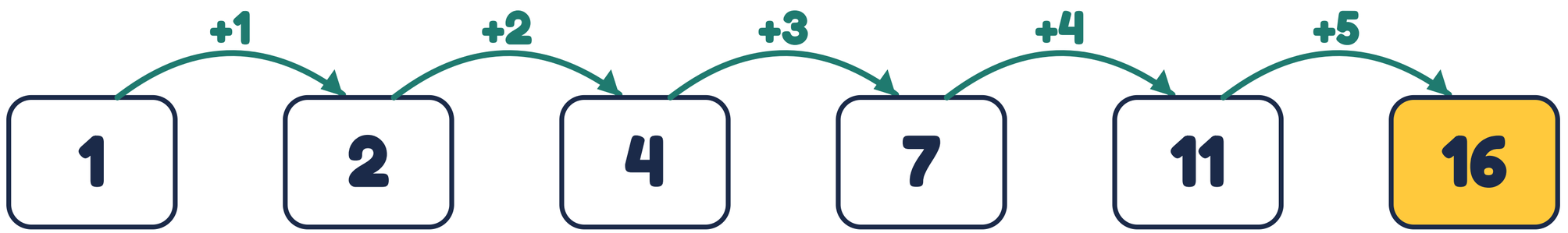
Counting in 10s: every number ends in 0.

Counting in 5s: every number ends in 0 or 5.

Counting in 2s: every number ends in 0, 2, 4, 6 or 8.

28–38 MIN

# Gaps can grow




The gap grows by 1 each time. So the next gap is 5.

28–38 MIN

# Quick-fire: 10 more

**23** 47 **65** 90

Say the number that is 10 more. Think: one row down.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

38–42 MIN

# Jump, clap, tap

A big jump for every ten. A clap for every five. A tiny tap for every two.

10 5 2

Sip some water.

42–70 MIN

# Workbook page 1: Start here

Questions 1 and 2. Timer: 8 minutes

**S1** Colour 10, 20, 30 and so on to 100 as you count in 10s. Fill in the blanks.

**S2** Count in 5s. Colour 5, 10, 15 and keep going to 100. Fill in the blanks.

42–70 MIN

# Check page 1

| | |
|---|---|
| S1 | They end in 0. They make a line going down. |
| S2 | They end in 5 or 0. They make 2 lines going down. |

42–70 MIN

# Workbook page 2: Start here

Questions 3 to 6. Timer: 10 minutes

**S3** What are the next two numbers in each of these patterns?

**S4** Count in 2s. Colour 2, 4, 6 and so on to 30. Look at the last digits.

**S5** Fill in the missing number in each empty square.

**S6** Siti counts in 5s: 5, 10, 15, 20, 26, 30, 35. One number is wrong. Circle it.

# Check page 2

42–70 MIN

| | |
|---|---|
| S3 | 10, 12 \| 30, 35 \| 60, 70 \| 41, 51 |
| S4 | 2, 4, 6, 8, 0 (any order) |
| S5 | a 34, b 43, c 55 |
| S6 | 26 is wrong. She should have said 25. Counting in 5s ends in 0 or 5. |

42–70 MIN

# Choose your next page

| Page | What it is |
| --- | --- |
| **Think harder (page 3)** | Our idea in a new setting |
| **Go further (page 4)** | Bigger puzzles |
| **Investigation (page 5)** | Counting in 3s |
| **Check** | Finish and check your own work |

**Everyone may choose. You can swap at any time.**

# Talk and explain

70–76 MIN

ALI SAYS

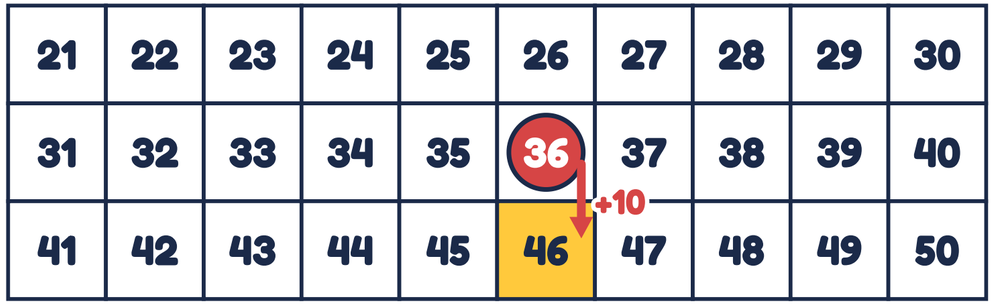
Counting in 5s makes one column on the square, just like counting in 10s.

Is Ali right? Use the 100 square to show why.

I think ... because ...

# Back to the sneak-peek

70–76 MIN




**The counter is on 46. Down one row is 10 more.**

# Exit ticket

76–82 MIN

On your own. No looking back. Timer: 3 minutes

**E1** Count in 5s. What are the next two numbers?

**E2** A counter is on 26. First it moves down 1 square. Then it moves right 1 square.

**E3** Mei counts in 2s: 2, 4, 6, 8 and so on. Will she say 25?

# Check the exit ticket

76–82 MIN

| | |
|---|---|
| E1 | 50, 55 |
| E2 | 36, then 37 |
| E3 | No. 25 ends in 5. Counting in 2s ends in 0, 2, 4, 6 or 8. |

# Look-back

82–85 MIN

**Three minutes. There are no right answers. You can pass. Nobody reads it out.**

| | |
|---|---|
| **Something I did well** | write or draw one thing |
| **The hardest part was …** | ☐ finding my place on the square ☐ counting in 2s, 5s or 10s ☐ moving down a row<br>☐ the pattern where the gap grows (Think harder) ☐ something else |
| **What helped me** | ☐ the 100 square or counters ☐ drawing it ☐ trying again ☐ I asked for help<br>☐ I was not stuck |
| **Next time I will try** | ☐ drawing it ☐ using counters ☐ checking my answer ☐ asking a question |
| | ☐ I am still stuck (my tutor will come and help) ☐ I would like to pass |

**Take your time. You can pass.**



85–90 MIN

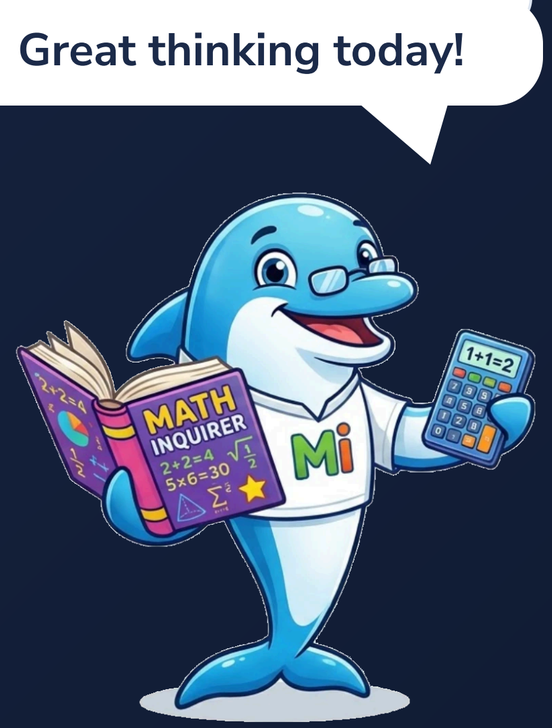
# Well done, inquirers!

Show a grown-up one thing you found out today.

NEXT LESSON

Adding by making tens

You need a pencil and paper.



OPTIONAL

# Go further: what if?

| | | | | |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 |
| 16 | 17 | 18 | 19 | 20 |

Here is a square with only 5 numbers in each row.

A counter is on 8. It moves down 1 row. Which number is it on?

What does going down add now?

OPTIONAL

# If we have time: pick one

| Activity | What we do |
| --- | --- |
| **Is it always true?** | Add 10 to a number. Does only the tens digit change? |
| **Make a puzzle** | Hide a number on the square and write clues for your grown-up |
| **Teach it back** | Show why counting in 10s makes a column |

**Optional. Everyone may join.**